COMP 2087: Programming for Beginners II

Assignment #2

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For each question, write a program **with comments** to perform the required tasks.
(Q1 = **58** points; Q2 = **27** points; Q3 = **24** points. TOTAL = **109** points)

1.
   a. Write a function *encrypt()* that takes a str ***s1*** as input and creates a new str ***s2*** by subtracting 9 from the ascii code for each character in the string ***s1***.
   b. Write a function *decrypt()* that takes a str ***s1*** as input and creates a new str ***s2*** by adding 9 to the ascii code for each character in the string ***s1***.
   c. Read each line of a file *infileA2Q1.txt* and pass the line (after removing any whitespace characters at the end) to the function *encrypt()*. The output of *encrypt()* should be written on a new line in the output file *encrypted.txt*
   d. Read each line of a file *encrypted.txt* and pass the line (after removing any whitespace characters at the end) to the function *decrypt()*. The output of *decrypt()* should be written on a new line in the output file *decrypted.txt*
   e. On the screen, print the total number of (non-empty) lines in *encrypted.txt*

2. Write a program that reads the contents of a file *infileA2Q2.txt* and creates a dictionary ***d1*** where the keys are individual words in the file. The value corresponding to a key is the number of times the word appears in the file. For each item, in the ***d1***, print the key and corresponding value with a suitable message. Also print the total number of unique words in the file.

   Sample output (on screen portion only):

```
"C:\Program Files (x86)\Python312\python.exe" "C:\U
The word "How" occurs 1 time(s) in the file.
The word "much" occurs 1 time(s) in the file.
The word "wood" occurs 3 time(s) in the file.
The word "would" occurs 1 time(s) in the file.
The word "a" occurs 3 time(s) in the file.
The word "woodchuck" occurs 3 time(s) in the file.
The word "chuck" occurs 3 time(s) in the file.
The word "if" occurs 1 time(s) in the file.
The word "could" occurs 1 time(s) in the file.
The word "But" occurs 1 time(s) in the file.
The word "can't" occurs 1 time(s) in the file.
No. of unique words in the file is: 11
```

3. Define a list **list1 = [12, 35.8, -9, 23, -18.0, 54]**. Next, for each line of the file *infileA2Q3.txt* do the following:

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a. Remove any whitespace characters at the end of the line and convert to an integer ***n*** and print the value of ***n***
b. If ***n*** > 10 set ***i*** = 0.25***n***; otherwise set ***i*** = ***n***
c. Print list1[***i***] with a suitable message.
d. Print list1[0]//***i*** with a suitable message.

If any exceptions occur, print the default error message for the exception. For *ValueError* also print a custom message in addition to the default error message, then continue with the next item in the file.

Sample output:

```
"C:\Users\12262\OneDrive - University of Windsor\208
projects\Assignments\F24assn2.py"
******
n = 2 ;  i = 2
list1[2] = -9
list1[0]//2 = 6
******
n = -1 ;  i = -1
list1[-1] = 54
list1[0]//-1 = -12
******
invalid literal for int() with base 10: '7.5'
Value on line could not be converted to int
******
n = 1 ;  i = 1
list1[1] = 35.8
list1[0]//1 = 12
******
n = 16 ;  i = 4.0
list indices must be integers or slices, not float
******
n = 0 ;  i = 0
list1[0] = 12
integer division or modulo by zero
******
n = 9 ;  i = 9
list index out of range
******
n = 2 ;  i = 2
list1[2] = -9
list1[0]//2 = 6
```